
KTU
NOTES

MODULE II

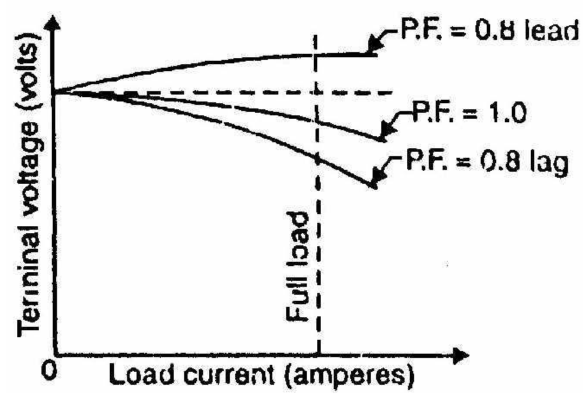
- Voltage regulation
  - ✓ EMF
  - ✓ MMF
  - ✓ ASA
  - ✓ POTIER
- Blondel's two reaction theory
- Parallel operation of alternators

Voltage regulation

- Voltage regulation of an alternator is defined as the increase in terminal voltage expressed as the percentage of rated terminal voltage, when load at a given power factor is thrown off with speed & field excitation remains constant

i.e, $$\% Voltage \ \ regulation = \frac{E_0 - V}{V} * 100\%$$

- Where, E_0 = No load terminal voltage, V = Full load terminal voltage
- When load on Alternator changes, its terminal voltage also changes
- The magnitude of this change depends on load and load power factor
- The effect of change in load power factor on terminal voltage is shown in figure



- In the case of lagging pf, terminal voltage will rise & in the case of leading pf, terminal voltage will fall with removal of load

Determination of Voltage regulation

1. Direct method

- This method is used to find the regulation of small machines
- Here the Alternator is driven at synchronous speed & field excitation is adjusted to get rated terminal voltage (V)
- Now load is varied to get desired load at desired pf by keeping speed & terminal voltage (V) constant
- Then the entire load is thrown off by keeping the speed & field excitation constant to get no load terminal voltage (E_0)

Now,

$$\boxed{\%\,Voltage\ Regulation = \frac{E_0 - V}{V} * 100\%}$$

- The kVA rating of commercial Alternators are very high & the cost of finding the regulation by direct loading is high
- So voltage regulation of large Alternators are determined by Indirect method

2. Indirect method

The indirect methods commonly used to find the regulation of an Alternator are

1. **EMF method**
2. **MMF method**
3. **ZPF method**
4. **ASA method**

All these methods require the following data

i) Armature (stator) resistance

ii) Open Circuit Characteristics (OCC)

iii) Short Circuit Characteristics (SCC)

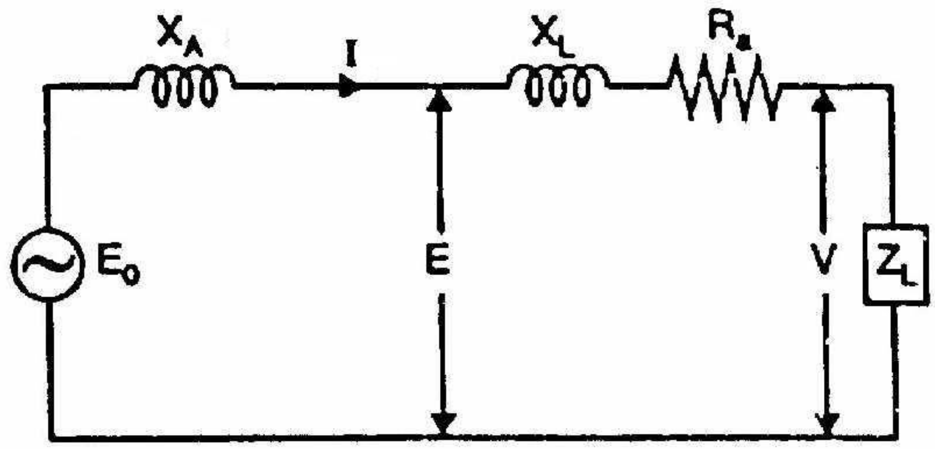
1. EMF method or Synchronous impedance method

- In this method of finding the voltage regulation of an alternator, we find the synchronous impedance Z_S (and hence synchronous reactance X_S) of the alternator from the OCC and SCC
- For this reason, it is called synchronous impedance method

- When load on the alternator is varied (i.e, armature current is varied), the terminal voltage (V) of the alternator also varies, even if the speed & field excitation kept constant

- The variation in terminal voltage is due to the following reasons

1. Voltage drop due to armature resistance (IR_a drop)
2. Voltage drop due to armature leakage reactance (IX_L drop)
3. Voltage drop due to armature reaction (IXa)



Here, E_0 = No load EMF

E = Induced EMF after allowing armature reaction

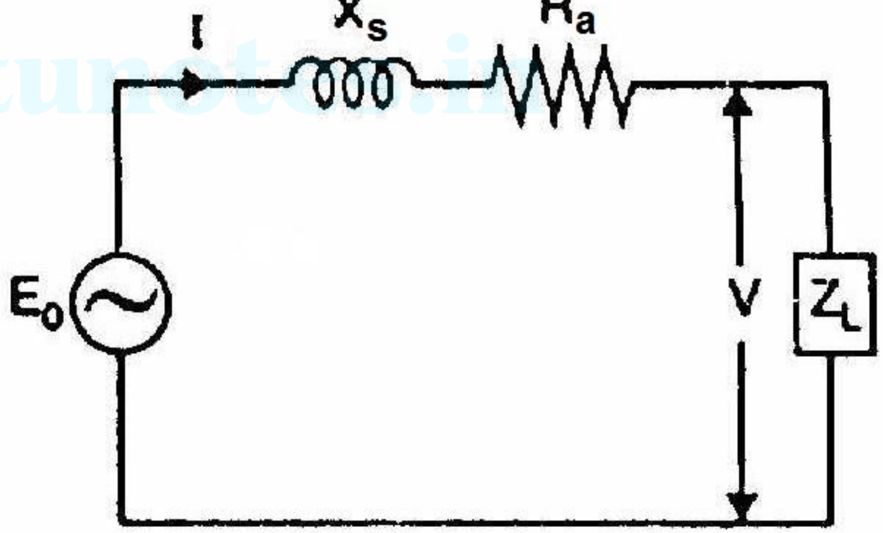
V = Terminal voltage

- $E = V + I(R_a + jX_L)$
- $E_0 = E + I(jX_a)$

Synchronous Impedance (Z_S)

- Synchronous impedance, $Z_S = R_a + j X_S$
- The synchronous impedance is the fictitious impedance employed to account for the voltage effects in the armature circuit produced by the actual armature resistance, the actual armature leakage reactance and the change in the air gap flux produced by armature reaction.
- $E_0 = V + IZ_S$

$$= V + I(R_a + jX_S)$$

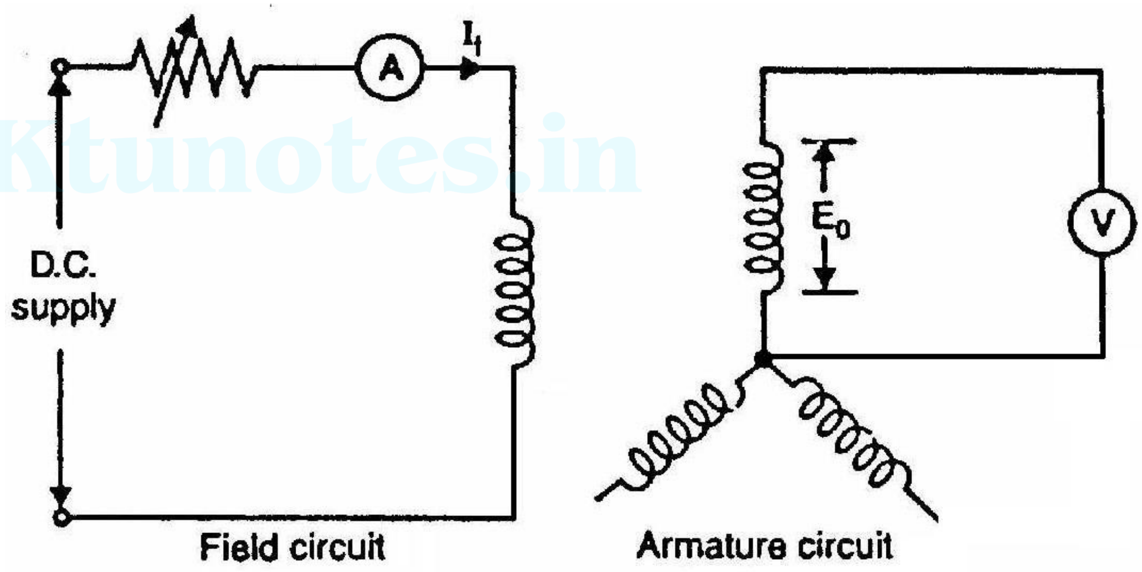


Determination of Synchronous Impedance, Z_S

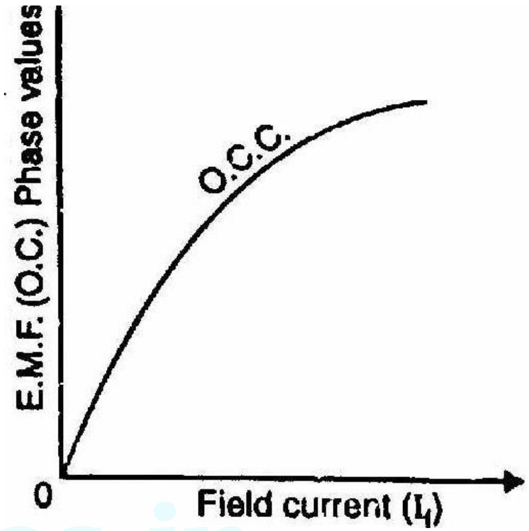
Synchronous impedance is determined from open circuit & short circuit tests

a) Open circuit test

The connection diagram for open circuit test is shown in figure.



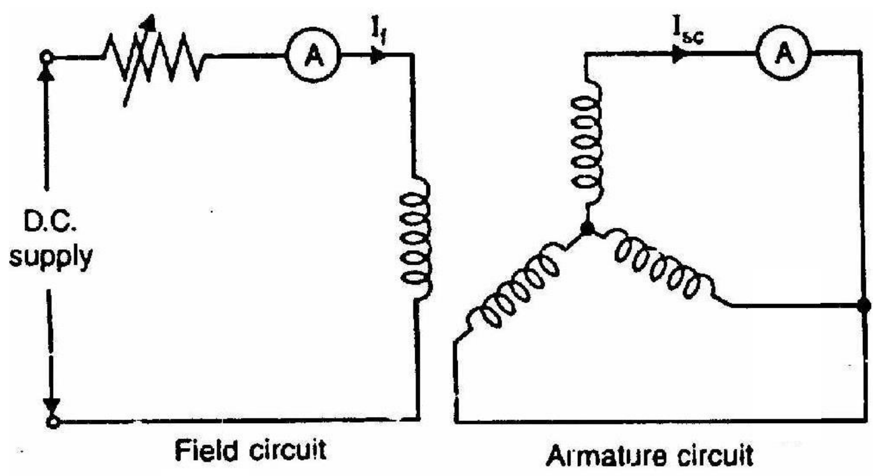
- The alternator is run on no load at the rated speed. The field current I_f is gradually increased from zero (by adjusting field rheostat) until open circuit voltage E_0 (phase value) is about 50% greater than the rated phase voltage.

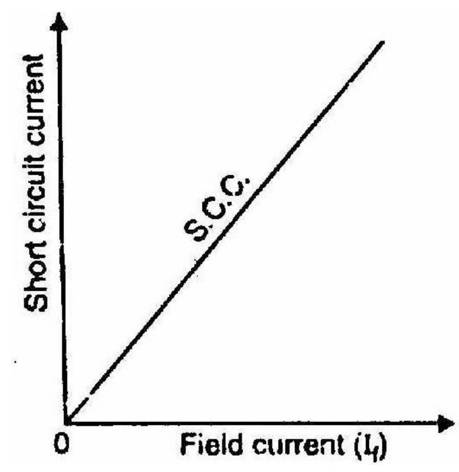


- The graph is drawn between open circuit voltage values and the corresponding values of I_f as shown
- The curve drawn between open circuit voltage and corresponding field current is called Open Circuit Characteristics (OCC)

b) Short circuit test

- In a short circuit test, the alternator is run at rated speed and the armature terminals are short circuited
- The field current I_f is gradually increased from zero until the short circuit armature current I_{SC} is about the rated current.





- The graph between short circuit armature current & corresponding field current is called Short Circuit Characteristics (SCC)
- SCC is normally a straight line through origin because the net excitation required is so small as there is no saturation in magnetic circuit

<u>Determination of synchronous impedance from OCC & SCC</u>

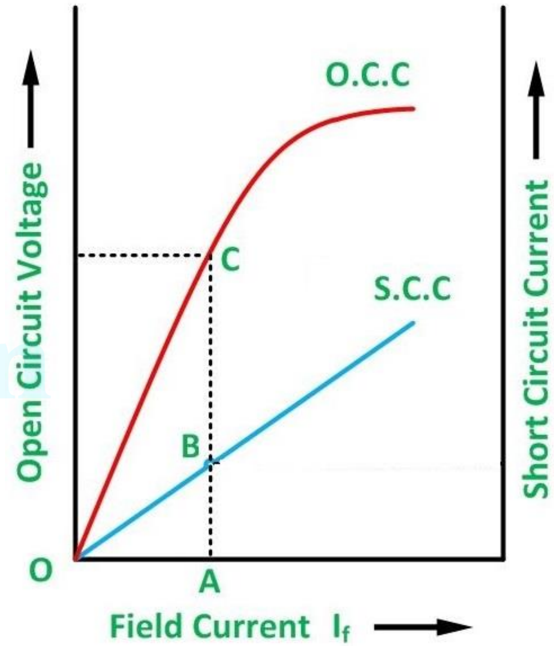
- Synchronous impedance can be determined by knowing the value of short circuit current I_{SC} corresponding to a field current that gives rated terminal voltage/phase on open circuit

- Synchronous Impedance = $\dfrac{\text{Open circuit voltage}}{\text{Short circuit current corresponding to field current which gives rated voltage/phase}}$
- If OA is the field current that gives the rated EMF/phase represented by AC & AB gives the short circuit current I_{SC} corresponding to field current OA
- On short circuit, the terminal voltage is zero & the whole of induced EMF is utilized to create a short circuit current (I_{SC}) through synchronous impedance (Z_S)

  i.e, $E_0 = I_{SC} Z_S$

  Synchronous impedance, $Z_S = E_0/I_{SC}$

  $= AC/AB$



The armature resistance can be found by ammeter-voltmeter method.

$$\therefore \quad X_S = \sqrt{Z_S^{\ 2} - R_a^{\ 2}}$$

Once we know R_a and X_S, the phasor diagram can be drawn for any load and any pf.

- The phasor diagram for the inductive load is shown in figure; the load pf being cosϕ lagging. Note that in drawing the phasor diagram, current 'I' has been taken as the reference phasor. The IR_a drop is in phase with 'I' while IX_S drop leads 'I' by 90°. The phasor sum of V, IR_a and IX_S gives the no load EMF E
- Consider the triangle OBD.

OB = VCosϕ + IR_a

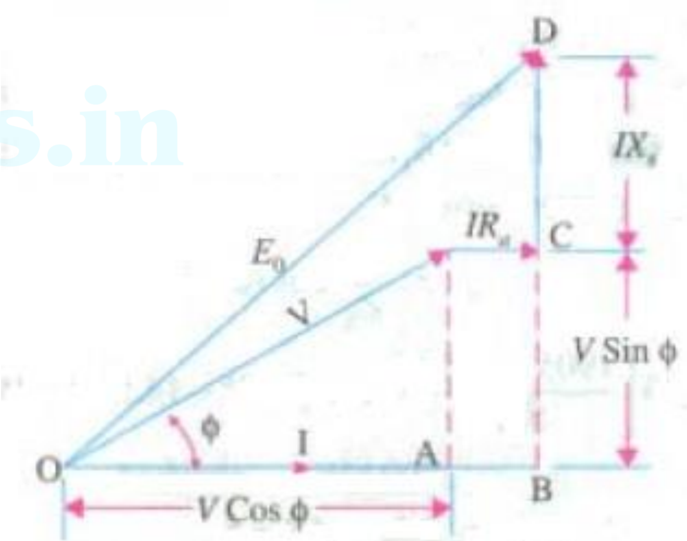
BD = VSinϕ + IX_S

OD = E_0,

$$\therefore E_0 = \sqrt{OB^2 + BD^2}$$

$$i.e, \quad E_0 = \sqrt{(VCos\phi + IR_a)^2 + (VSin\phi + IX_s)^2}$$

$$\%\,Voltage\ regulation = \frac{E_0 - V}{V} * 100\%$$



- For leading pf, take ϕ as negative & for unity pf, take ϕ as zero
- This method is called pessimistic method because the regulation obtained by this method is always higher than actual value.
- The reason for error is that here synchronous impedance is assumed to be constant while actually it is not.
- Synchronous impedance varies with saturation.

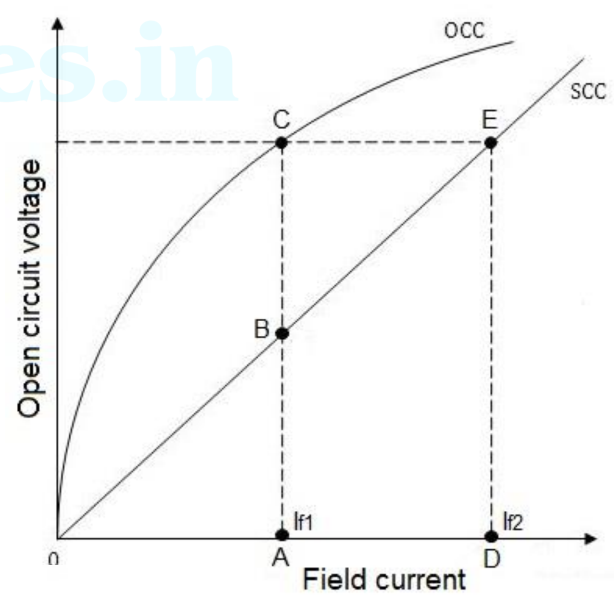
Short circuit ratio (SCR)

- The SCR of a synchronous machine is defined as the ratio of field current to produce rated voltage on open circuit to field current required to circulate rated current on short circuit while the machine is driven at synchronous speed

$$SCR = \frac{Field\ Current\ required\ to\ produce\ rated\ Voltage\ on\ open\ circuit}{Field\ current\ required\ to\ produce\ rated\ Current\ on\ short\ circuit}$$

From graph,

$$SCR = \frac{I_{f1}}{I_{f2}} = \frac{OA}{OD} = \frac{AB}{DE} = \frac{AB}{AC} = \frac{1}{AC/AB}$$



- AC/AB = X_S

  So **SCR = 1/Xs**
- A small value of SCR indicates small value of short circuit current owing to large value of synchronous reactance.
- As SCR increases, stability limit increases & voltage regulation improves.

2. MMF or Ampere Turn method

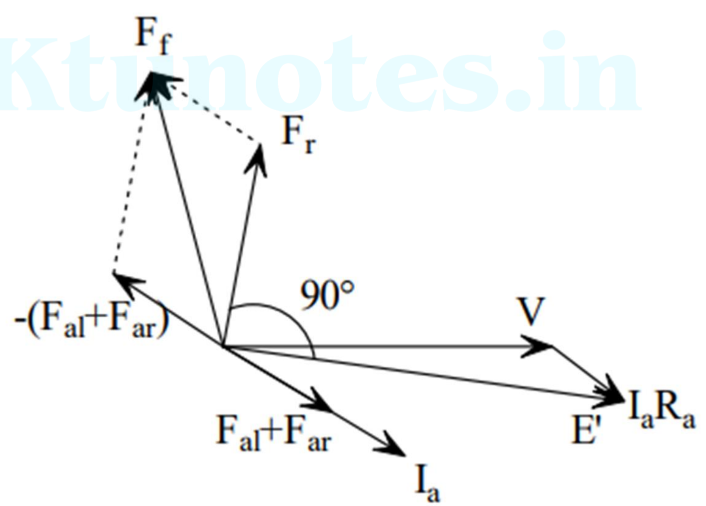
- In this method, the armature leakage reactance is considered as an additional armature reaction
- Here it is assumed that the change in terminal voltage on load is entirely due to armature reaction (voltage drop due to armature resistance is neglected)
- The field AT required to produce a voltage at full load is equal to the vector sum of field AT required to produce voltage 'V' at no load & field AT required to overcome the effect of armature reaction
- Here EMFs are replaced by MMFs (field MMF = NI_f , N = number of turns in field coil & I_f = field current. Number of turns is a constant & we can consider MMF α I_f)

MMF method

I_aR_a and I_aX_{al} are emf quantities and armature reaction is an mmf quantity. In emf method, all are treated as emf quantities (armature reaction mmf is replaced by I_aX_{ar}) while in mmf method, all of them are treated as mmf quantities.

Armature reaction mmf F_{ar} is in phase with I_a. Armature leakage reactance drop I_aX_{al} is transformed into equivalent mmf F_{al}. As F_{ar} is in phase with I_a, similarly F_{al} is also in phase with I_a.

Field mmf $\overline{F}_f = \overline{F}_r - (\overline{F}_{al} + \overline{F}_{ar})$ where $\overline{F}_r$ is the mmf corresponding to $\overline{E}' = \overline{V} + \overline{I}_a \times R_a$



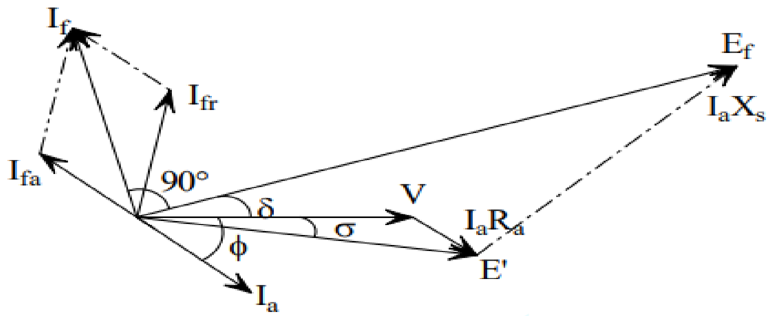
F_r (obtained from OCC corresponding to E') leads from E' by 90°.

$\overline{F}_{al} + \overline{F}_{ar}$ is the mmf (field current) required to circulate full-load armature current under short circuit test. $\overline{F}_{al} + \overline{F}_{ar}$ is in phase with the armature current I_a.

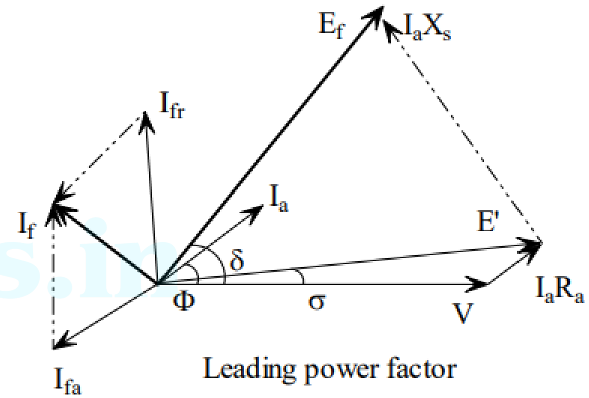
Steps to obtain voltage regulation by mmf method:

1) Plot O.C.C. & S.C.C.
2) Find $\overline{E}' = V\angle 0 + I_a\angle^{\pm}\Phi \times R_a = |E'|\angle^{\pm}\sigma°$ and obtain I_{fr} corresponding to E' from OCC. $\overline{I}_{fr} = I_{fr}\angle^{\pm}\sigma + 90°$ (Note: -Φ for lag & +Φ for lead)
3) Find I_{fa} from SCC. I_{fa} is the field current to circulate full-load armature current on short circuit (from SC test). $\overline{I}_{fa} = I_{fa}\angle^{-}\Phi + 180°$ for lag and $\overline{I}_{fa} = I_{fa}\angle^{+}\Phi + 180°$ for lead
4) Find $\overline{I}_f = \overline{I}_{fr} + \overline{I}_{fa} = I_f\angle 90 + \delta°$
5) Corresponding to I_f, obtain E_f from OCC.
6) Voltage regulation = $\dfrac{E_f - V}{V} \times 100\,\%$

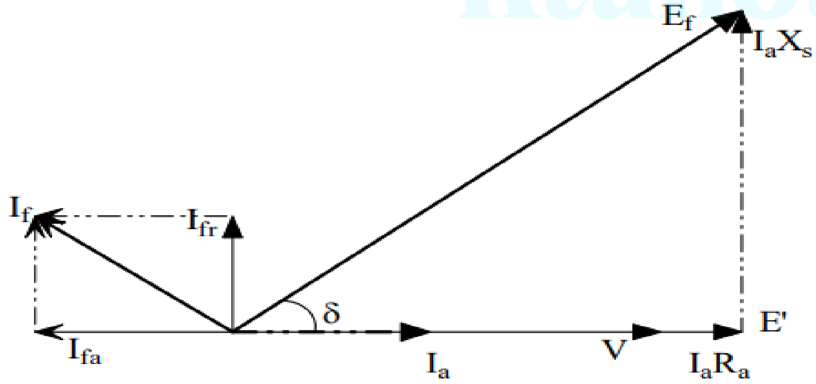
Phasor diagram of MMF method



Lagging p.f



Leading power factor



Unity power factor

Note : MMF methods gives low value of voltage regulation compared to actual value. Hence mmf method is also called optimistic method.

Q1: The open circuit and short circuit test is conducted on a 3-phase, star connected, 220V, 15.25kVA alternator. The OC test results are,

I_f amp	0.2	0.4	0.6	0.8	1	1.2	1.4	1.8	2.2	2.6
E_f line volts	29	58	87	116	146	172	194	232	262	284

Field current of 1.2A produces a short circuit current of 40A. R_a = 0.06Ω/phase. Calculate its full load regulation at 0.8 p.f. lag using mmf method.

Ans:

Full load current, $I_a = \frac{15250}{\sqrt{3}\times 220} = 40A$ $\quad$ $V = \frac{220}{\sqrt{3}} = 127V$

$\bar{E}' = V\angle 0 + I_a\angle^{-}\Phi \times R_a = 127\angle 0 + 40\angle^{-}36.87 \times 0.06 = 128.9\angle^{-}0.64$ V

$I_{fr} = 1.7A$ (from OCC corresponding to E_f = 128.9V)

$\bar{I}_{fr} = I_{fr}\angle -\sigma + 90° = 1.7\angle 89.36°$

I_{fa} = 1.2A (from SCC corresponding to I_{sc} = 40A)

$\bar{I}_{fa} = I_{fa}\angle^{-}\Phi + 180° = 1.2\angle 143.13°$ A

$\bar{I}_f = \bar{I}_{fr} + \bar{I}_{fa} = 1.7\angle 89.36° + 1.2\angle 143.13° = 2.6\angle 111.25°$ A

E_f = 165V (from OCC corresponding to I_f = 2.6A)

Voltage regulation = $\frac{165-127}{127} \times 100\% = 29.92\%$

3. Zero Power Factor (ZPF) or Potier Method

- The voltage regulation obtained by EMF & MMF methods is based on the total synchronous reactance
- This method is based on the separation of reactance into leakage reactance & reactance due to armature reaction. Therefore this method is more accurate
- The data required for this method are

i) OCC ii) effective armature resistance iii) magnitude of field current required to circulate full load current in stator iv) Zero Power Factor (ZPF) full load voltage characteristics

- ***Plotting ZPF curve***
- This is the curve between terminal voltage & field current, when the alternator is delivering full load current to a zero pf (lagging) load
- The test is carried out by running the alternator at synchronous speed & connecting a purely inductive 3ϕ load to its terminals
- There is no need of plotting the full curve, only points A & B are sufficient.
- Point ‘B’ corresponds to the field current which gives rated terminal voltage, when the machine is delivering full load current at zero pf

- Point ‘A’ corresponds to the field current required to circulate rated current on short circuit (from short circuit test. Since armature resistance is very small, during short circuit, the circuit can be considered as purely inductive)

Procedure for Potier method

Step 1 - Plot OCC

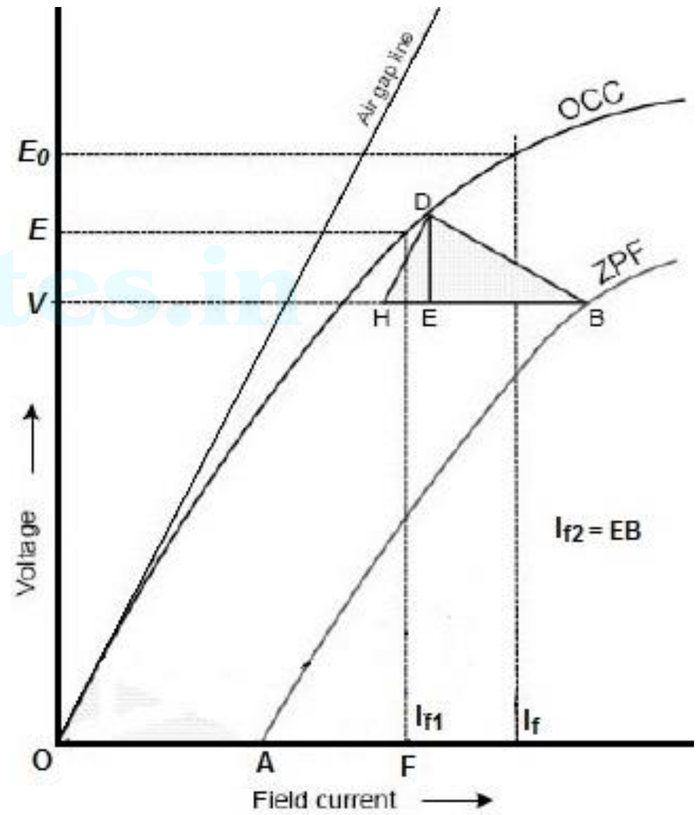
Step 2 – Find armature resistance (R_a)

Step 3 – Plot ZPF curve

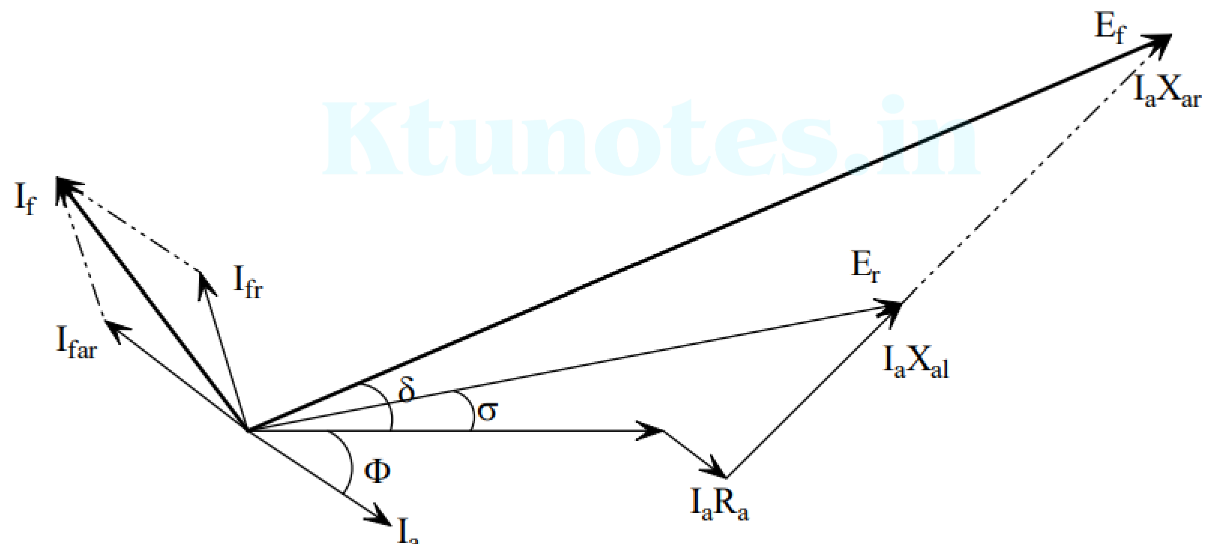
Step 4 – Draw a line tangent to OCC, which is known as air gap line

Step 5 – Find the point corresponding to V (rated terminal voltage) on ZPF curve (i.e, point B)

Step 6 – Draw the triangle BHD such that BH = OA & HD parallel to air gap line, D is the point on OCC



- Complete the triangle BHD. From ‘D’ draw the line DE perpendicular to BH. The ***triangle BED is known as Potier triangle***.
- ***The length DE represents IX_L drop & HE represents field current corresponding to leakage reactance drop.***
- ***EB represents field current corresponding to armature reaction drop***

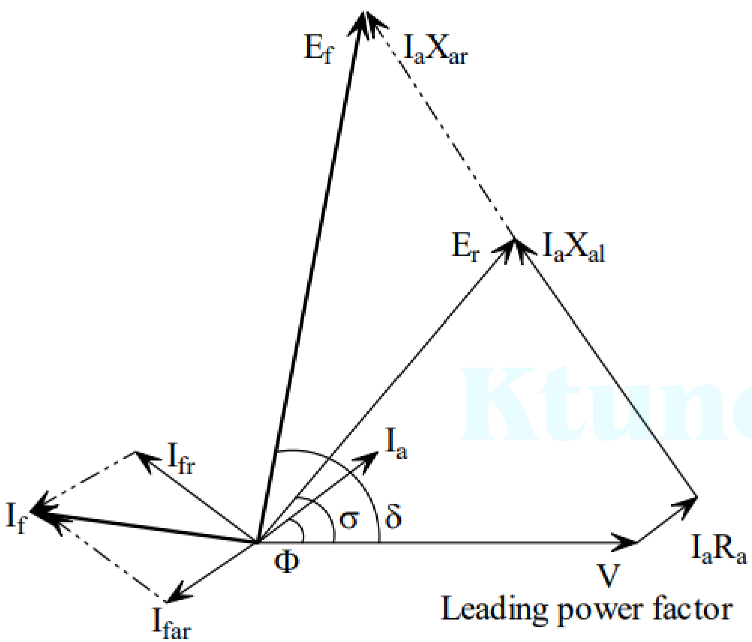


Lagging power factor

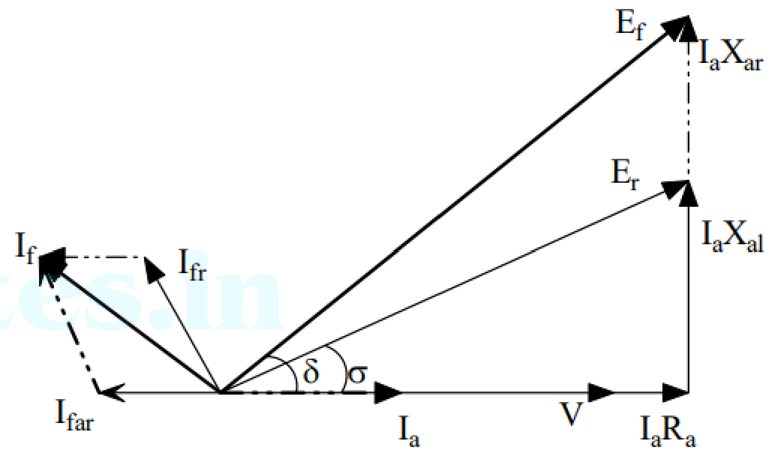
Steps to obtain voltage regulation by potier method:

1) Plot O.C.C.
2) To plot Z.P.F.C., obtain the points F & A. Point F corresponds to full load armature current during short circuit. Point A corresponds to rated terminal voltage and rated armature current when the load is zero power factor lagging. Draw Potier triangle. Find X_{al}.
3) Find $\bar{E}_r = V\angle 0 + I_a\angle^{\pm}\Phi \times (R_a + jX_{al}) = E_r\angle\sigma°$. Refer OCC and find I_{fr} corresponding to E_r. $\bar{I}_{fr} = I_{fr}\angle 90+\sigma°$
4) Find I_{far}. $\bar{I}_{far} = I_{far}\angle 180-\Phi°$ for lag $\bar{I}_{far} = I_{far}\angle 180+\Phi°$ for lead.
5) Find $\bar{I}_f = \bar{I}_{fr} + \bar{I}_{far} = I_f\angle 90+\delta°$
6) Refer OCC and find E_f corresponding to I_f. $\bar{E}_f = E_f\angle\delta°$
7) Voltage Regulation $= \dfrac{E_f - V}{V} \times 100\,\%$

Phasor diagram of ZPF method



Leading power factor



Unity power factor

Q1: The following data were obtained for the OCC of a 10MVA, 13kV, 3-phase, 50 Hz, star-connected synchronous generator:

I_f (A) :	50	75	100	125	150	162.5	200	250	300
V_{oc}(line)(kV)	6.2	8.7	10.5	11.8	12.8	13.2	14.2	15.2	15.9

An excitation of 100A causes the full-load current to flow during the shrot-circuit test. The excitation required to give the rated current at zero p.f. and rated voltage is 290A. Find the voltage regulation at full load and 0.8 p.f lag by a) emf method b) mmf method and c) potier method. Neglect armature resistance.

a) emf method

Full load current, $I_a = \dfrac{10\times10^6}{\sqrt{3}\times13000} = 444.1A$

$V = \dfrac{13000}{\sqrt{3}} = 7505.6V$

From OCC, corresponding to rated voltage (=7505.6V), $I_f = 155A$, $I_{sc} = 700A$;

$Z_s = \dfrac{7505.6}{700} = 10.72\Omega$

$X_s = 10.72\Omega$

$\Phi = 36.87°$

$E_f = V + I_a(R_a + jX_s) = 7505.6\angle 0° + 444.1\angle -36.87° \times j10.72 = 11039.8\angle 20.2° V$

V.R. = $\frac{11039.8 - 7505.6}{7505.6} \times 100 = 47.1\%$

b) mmf method

$\bar{E}' = V\angle 0 + I_a\angle^{-}\Phi \times R_a = 7505.6\angle 0\,\text{V}$

$I_{fr} = 155A$ (from OCC corresponding to E' = 7505.6V)

$\bar{I}_{fr} = I_{fr}\angle -\sigma + 90° = 1.55\angle 90°$

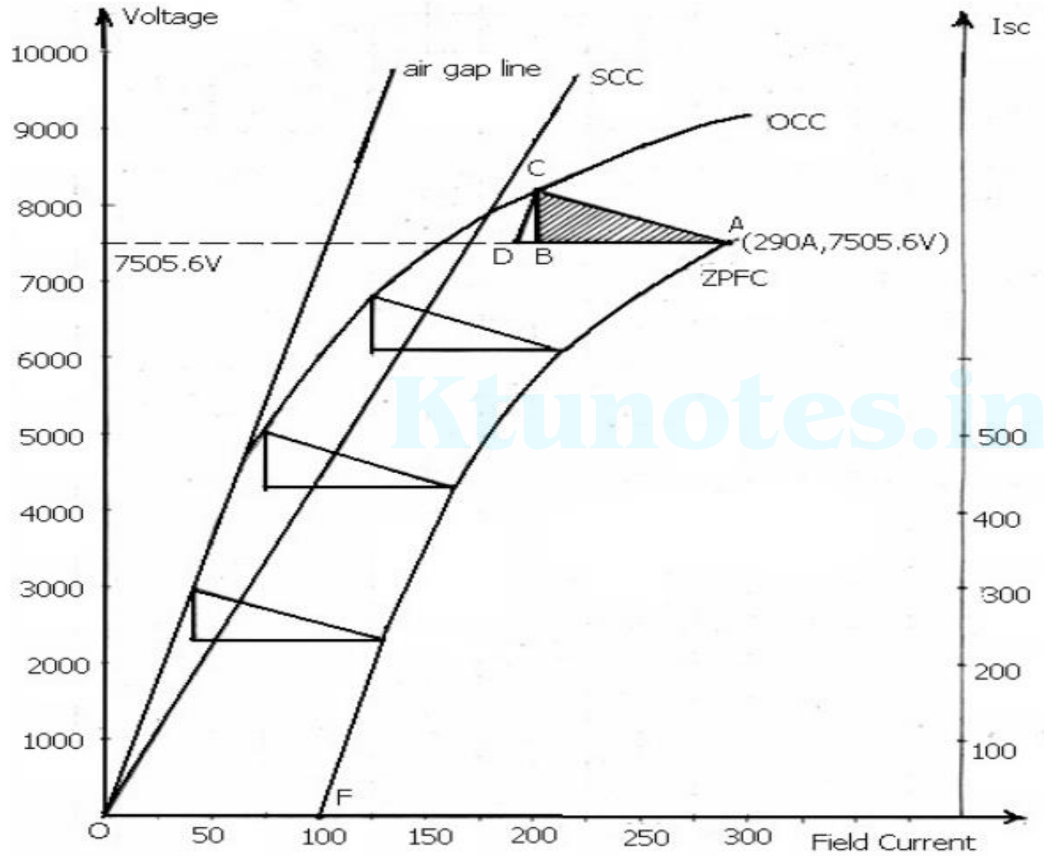
I_{fa} = 100A (from SCC corresponding to I_{sc} = 444.1A)

$\bar{I}_{fa} = I_{fa}\angle^{-}\Phi + 180° = 100\angle 143.13°\,\text{A}$

$\bar{I}_f = \bar{I}_{fr} + \bar{I}_{fa} = 155\angle 90° + 100\angle 143.13° = 229.4\angle 110.4°\,\text{A}$

E_f = 8500V (from OCC corresponding to I_f = 229.4A)

Voltage regulation = $\frac{8500 - 7505.6}{7505.6} \times 100 = 13.25\%$


Voltage
10000
9000
8000
7000
6000
5000
4000
3000
2000
1000
O
50
100
150
200
250
300
Field Current
Isc
500
400
300
200
100
air gap line
SCC
OCC
C
A
(290A,7505.6V)
D
B
7505.6V
ZPFC
F

c) POTIER method

$$X_{al} = \frac{BC}{I_a} = \frac{1.4 \times 500}{444.1} = 1.58\Omega$$

$$\overline{E}_r = V\angle 0 + I_a\angle^{-}\Phi \times (R_a + jX_{al}) = 7505.6\angle 0° + 444.1\angle -36.87° \times j1.58$$

$$= 7505.6\angle 0° + 444.1\angle -36.87° \times j1.58 = 7946.46\angle 4.05 = E_r\angle \sigma°$$

I_{fr} = 185A (from OCC corresponding to 7946.46V)

$$\bar{I}_{fr} = I_{fr}\angle 90 + \sigma° = 185\angle 94.05° \text{ A}$$

I_{far} = AB = 3.6*25 = 90A

$$\overline{I}_{far} = I_{far}\angle 180 - \Phi° = 90\angle 143.13° \text{ A}$$

$$\overline{I}_f = \overline{I}_{fr} + \overline{I}_{far} = 185\angle 94.05 + 90\angle 143.13 = 253.25\angle 109.6° = I_f\angle 90 + \delta°$$

E_f = 8800V (corresponding to I_f = 253.25A)

$$\bar{E}_f = 8800\angle 19.6°$$

$$\text{Voltage regulation} = \frac{8800 - 7505.6}{7505.6} \times 100\% = 17.25\%$$

D) ASA Modified MMF method

Neither of the two methods, MMF method and EMF method is capable of giving the reliable values of the voltage regulation due to the following reasons :

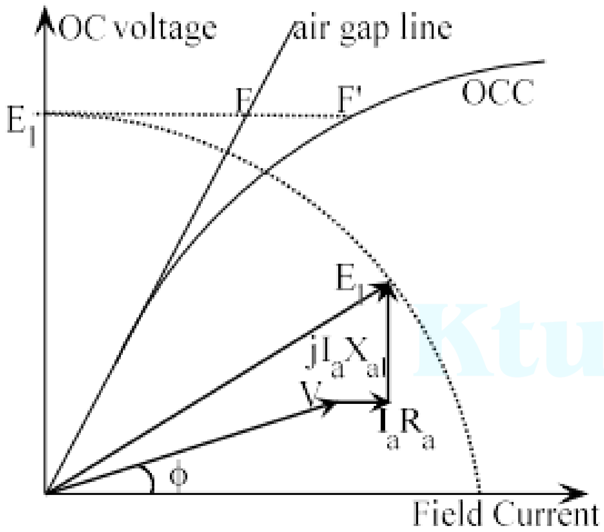
i) In these methods, the magnetic circuit is assumed to be unsaturated. This assumption is unrealistic as in practice. It is not possible to have completely unsaturated magnetic circuit.

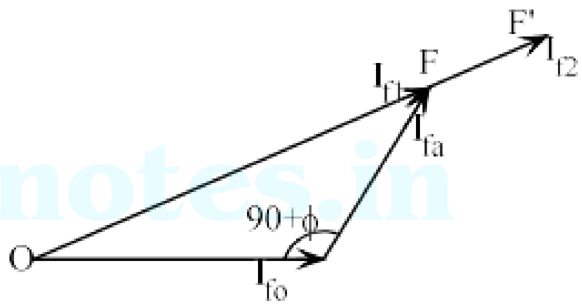
ii) In salient pole alternators, it is not correct to combine filed ampere turns and armature ampere turns. This is because the field winding is always concentrated on a pole core while the armature winding is always distributed. Similarly, the field and armature mmfs act on magnetic circuits having different reluctances in case of salient pole machine. Hence phasor combination field and armature mmf is not fully justified.

In ASA (American Standards Association) modified MMF method, additional excitation FF' required to compensate for the partial saturation of the machine is found.

I_{fo} is the excitation required to produce rated terminal voltage on open circuit and I_{fa} is the mmf required to produce rated armature current on short circuit (from SC test). ϕ is the power factor angle. Here, resultant OF (= I_{f1}) is based on the assumption of unsaturated magnetic circuit which is not true in practice. Additional mmf FF' is required to take into account the effect of partially saturated magnetic circuit.

From the Potier triangle, determine the leakage reactance X_{al} and find $\overline{E}_1 = V\angle 0 + I_a \angle^{-}\Phi \times (R_a + jX_{al})$. From E_1, draw a horizontal line intersecting air gap line at F and OCC at F'. FF' gives the additional excitation required to take into account the effect of partially saturated magnetic circuit. Adding FF' to I_{f1}, we get the total excitation I_{f2}. Corresponding to I_{f2}, we can determine the open circuit voltage E_f from OCC.





The results obtained by ASA method are reliable for both salient as well as non salient pole machines.

Q2: The following data were obtained for the OCC of a 10MVA, 13kV, 3-phase, 50 Hz, star-connected synchronous generator:

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V_{oc}(line)(kV)	6.2	8.7	10.5	11.8	12.8	13.2	14.2	15.2	15.9

An excitation of 100A causes the full-load current to flow during the shrot-circuit test. The excitation required to give the rated current at zero p.f. and rated voltage is 290A. Find the voltage regulation at full load and 0.8 p.f lag by ASA method. Neglect armature resistance.

Full load current, $I_a = \dfrac{10\times10^6}{\sqrt{3}\times13000} = 444.1A$

$V = \dfrac{13000}{\sqrt{3}} = 7505.6V$

$X_{al} = \dfrac{BC}{I_a} = \dfrac{1.4\times500}{444.1} = 1.58\Omega$

$\overline{E}_1 = V\angle0 + I_a\angle^{-}\Phi\times(R_a + jX_{al}) = 7505.6\angle0° + 444.1\angle-36.87°\times j1.58 = 7946.46\angle4.05$

Corresponding to E_1, distance between airgap line and OCC, FF'=70A

$I_{fo} = 155A$ (from OCC corresponding to V = 7505.6V)

I_{fa} = 100A

$\overline{I}_{f1} = \overline{I}_{fo} + \overline{I}_{far} = 155\angle0° + 100\angle(90-36.87)° = 229.4\angle20.4°A$

$I_{f2} = 229.4 + 70 = 299.4A$

E_f = 9180V (corresponding to 299.4A)

Voltage regulation = $\dfrac{9180-7505.6}{7505.6}\times100 = 22.3\%$

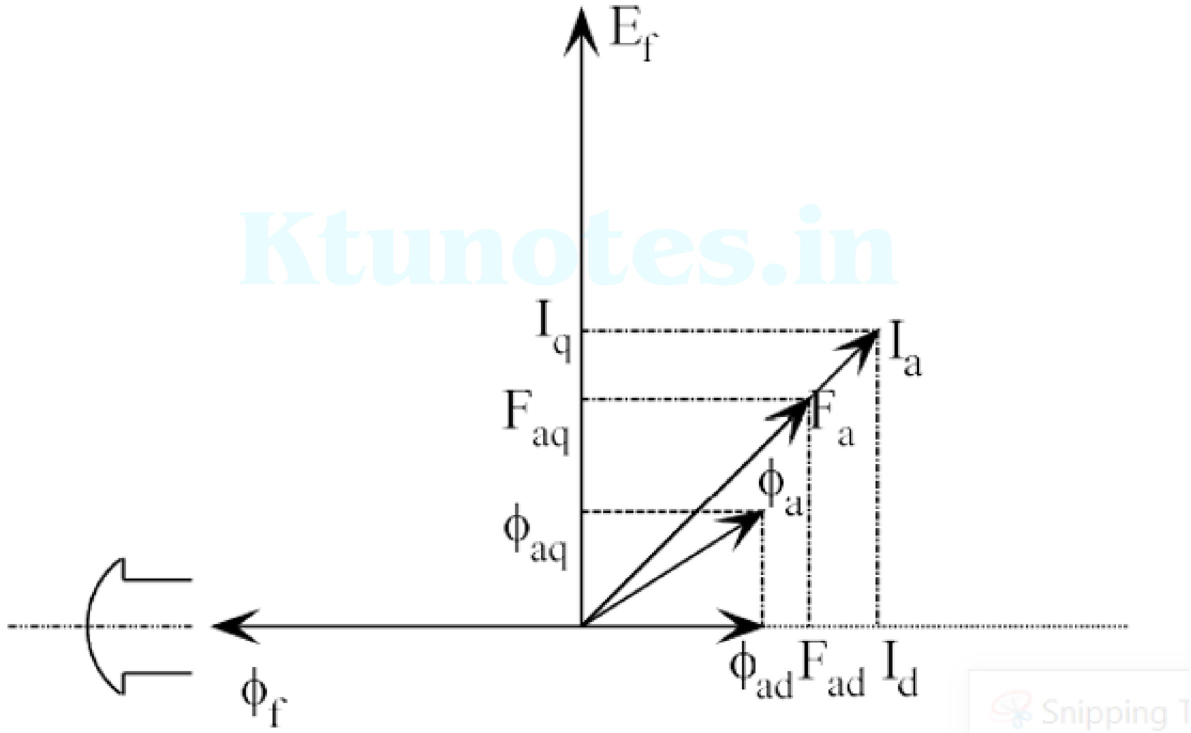
Note : voltage regulation by mmf method is 13.25% only.

BLONDEL'S TWO REACTION THEORY

- In a cylindrical rotor type machine, air gap is uniform so that the reluctance of the magnetic path is the same in all directions.
- Hence, effect of armature reaction can be accounted by one reactance Xs. Xs is constant for all directions of armature flux relative to rotor.

- However, in a salient pole type machine, the radial length of the air gap varies so that the reluctance of the magnetic circuit along polar axis (direct axis) is much less than the reluctance along the interpolar axis (quadrature axis).
- Because of the lower reluctance along the polar axis, more flux is produced along d-axis than along q-axis. Hence, reactance due to armature reaction will be different along d-axis and q-axis.
- According to two reaction theory, the armature mmf Fa can be divided into two components as
- component acting along direct axis Fad and
- component acting along quadrature axis Faq

Φf is the main flux and is along d-axis. Φf produces excitation emf Ef lagging by 90°. When generator is loaded, Ia flows. Assume that the power factor is lagging. Hence, Ia lags behind Ef. Ia creates armature mmf Fa. Fa can be decomposed into 2 components Fad and Faq. Fad produces flux Φad and Faq produces flux Φaq.



If Sd and Sq are the reluctances along direct and quadrature axes,

$$S_q > S_d$$

$$\frac{F_{aq}}{\Phi_{aq}} > \frac{F_{ad}}{\Phi_{ad}}$$

$$\frac{F_{aq}}{F_{ad}} > \frac{\Phi_{aq}}{\Phi_{ad}}$$

Resultant mmf, $F_a = F_{ad} + jF_{aq}$ and resultant flux, $\Phi_a = \Phi_{ad} + j\Phi_{aq}$

Φ_a is not in phase with armature mmf F_a but lags behind F_a.

Φ_{ad} will oppose the field flux Φ_f if the power factor is lagging.

Φ_{ad} will be in the same direction of the field flux Φ_f if the power factor is leading.

Φ_{aq} is cross magnetizing the main flux.

F_a is proportional to current I_a. Decomposing F_a into F_{ad} and F_{aq} in effect means decomposition of I_a into I_d and I_q.

Φ_{ad} and Φ_{aq} are rotating at synchronous speed. They induce armature reaction emfs in each phase of stator winding. These emfs are E_{ad} and E_{aq} respectively.

Resultant emf, $\bar{E}_r = \bar{E}_f + \bar{E}_{ad} + \bar{E}_{aq}$

Armature reaction emf can be considered as a voltage drop across a fictitious reactance. Hence,

$$\bar{E}_{ad} = -j\bar{I}_d X_{ad}$$

$$\bar{E}_{aq} = -j\bar{I}_q X_{aq}$$

$$\bar{E}_r = \bar{E}_f - j\bar{I}_d X_{ad} - j\bar{I}_q X_{aq}$$

Terminal voltage, $\bar{V} = \bar{E}_f - j\bar{I}_d X_{ad} - j\bar{I}_q X_{aq} - \bar{I}_a R_a - j\bar{I}_a X_{al}$

But, $\bar{I}_a = \bar{I}_d + \bar{I}_q$

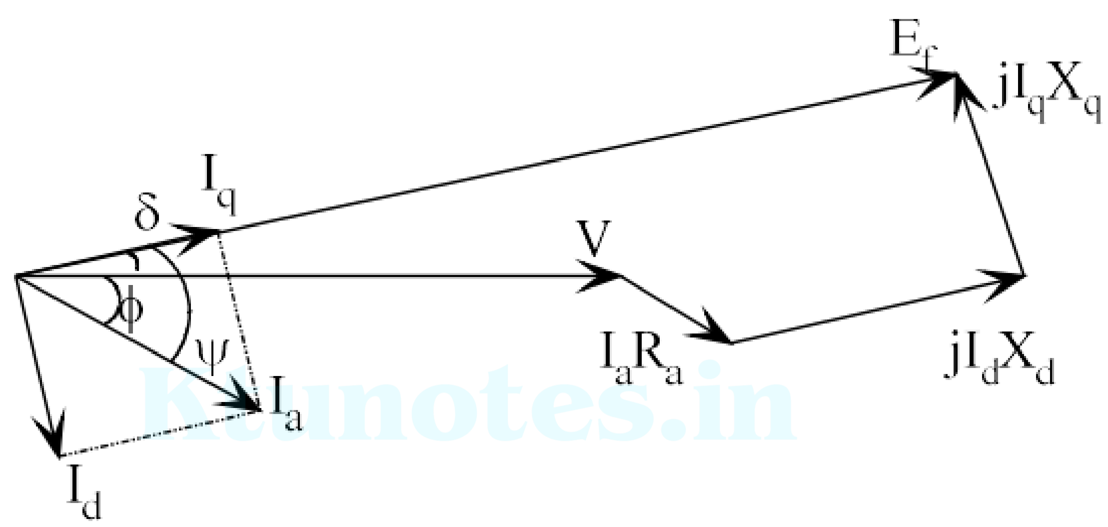
$$\begin{aligned}\bar{V} &= \bar{E}_f - j\bar{I}_d X_{ad} - j\bar{I}_q X_{aq} - \bar{I}_a R_a - j(\bar{I}_d + \bar{I}_q) X_{al} \\ &= \bar{E}_f - I_a R_a - j\bar{I}_d (X_{ad} + X_{al}) - j\bar{I}_q (X_{aq} + X_{al}) \\ &= \bar{E}_f - I_a R_a - j\bar{I}_d X_d - j\bar{I}_q X_q\end{aligned}$$

$X_d = X_{ad} + X_{al}$ = d-axis synchronous reactance

$X_q = X_{aq} + X_{al}$ = q-axis synchronous reactance

Note: X_{al} is independent of rotor position

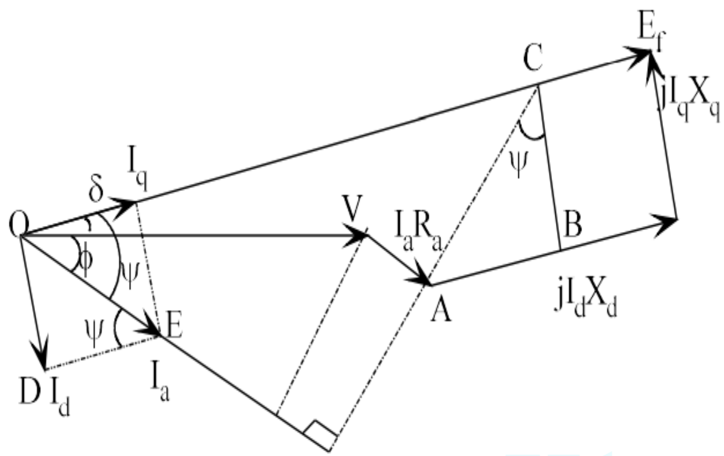
Note: I_q is along E_f.



$$I_q = I_a \cos \Psi$$

$$I_d = I_a \sin \Psi$$

Ψ = angle between I_a and E_f



ΔODE and ΔABC are similar. Hence, $\dfrac{AC}{OE} = \dfrac{BC}{DE}$

$$AC = \frac{BC \times OE}{DE} = \frac{jI_q X_q \times I_a}{I_q} = jI_a X_q$$

$$\vec{OC} = \bar{V} + \bar{I}_a R_a + j\bar{I}_a X_q = \bar{V} + \bar{I}_a (R_a + jX_q)$$

Angle of $\vec{OC}$ will give δ. (Taking V as reference)

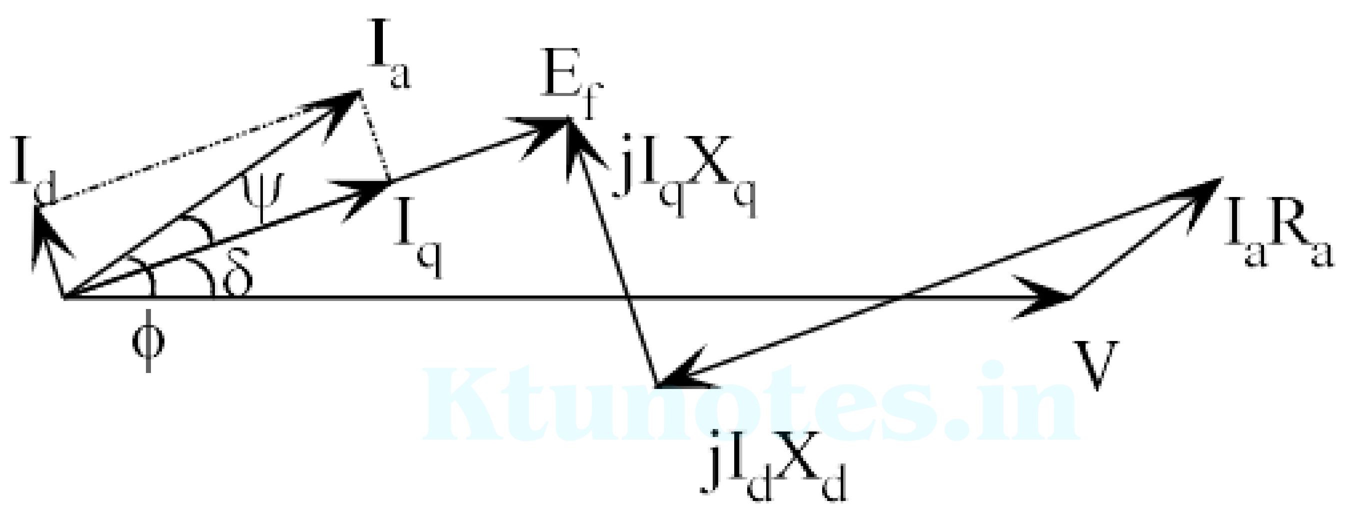
$\psi = \delta + \phi$ for lag

$\psi = \delta - \phi$ for lead

(or use the following formula for finding ψ,

$$\tan \Psi = \frac{V \sin\phi + I_a X_q}{V \cos\phi + I_a R_a})$$

$$E_f = V \cos\delta + I_a R_a \cos\Psi + I_a X_d \sin\Psi$$



Phasor diagaram at leading p.f.